

Problem Statement: Assist Ukrainian people in the rebuilding of their agricultural infrastructure



Conflict Map in Ukraine Burning of Wheat Field

Voice of the Customer

Important features:

- Affordability
- Durability
- Ease of use
- Efficiency

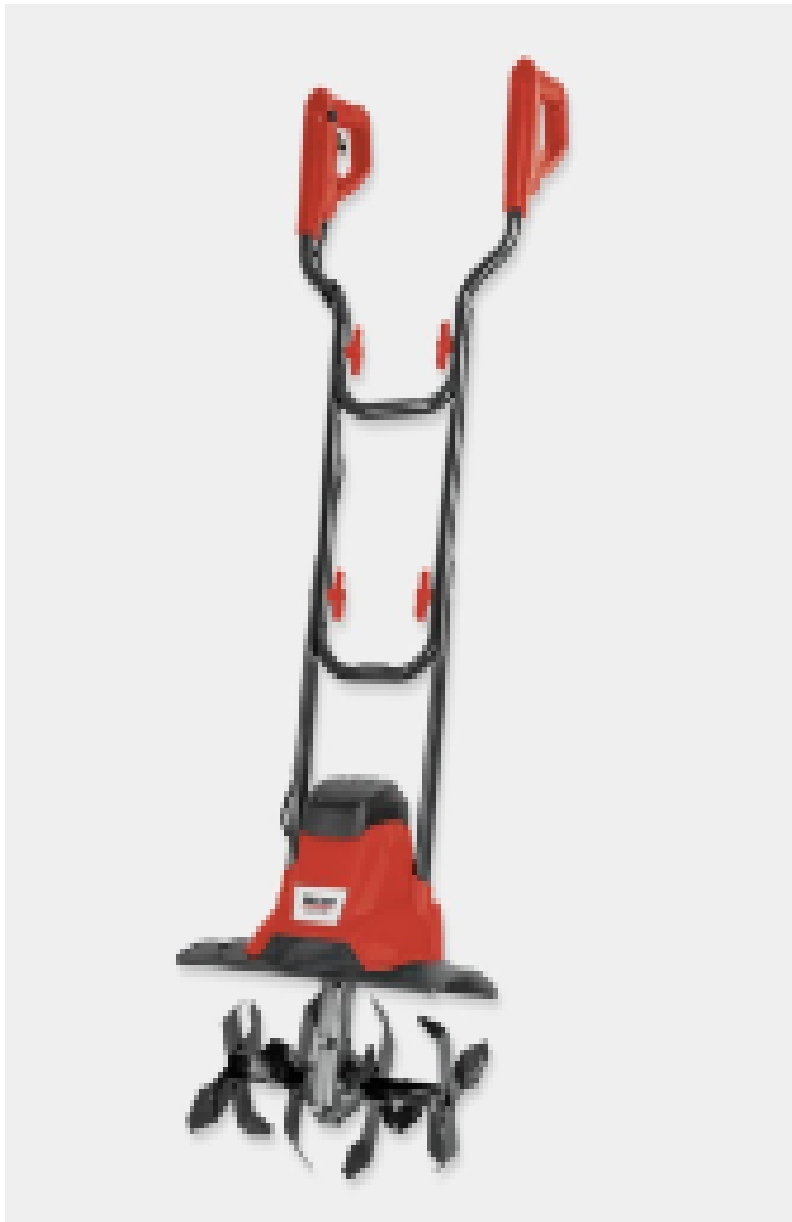
Specifications and Benchmarking

Chief competitor: Bodenackerfräse Electric Tiller and Cultivator 1200 W

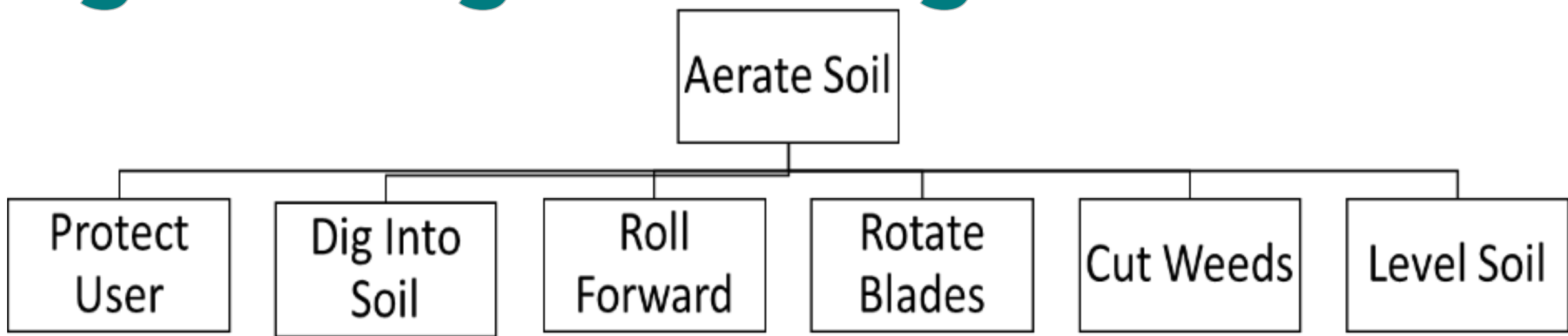
Top features:

- Easy to use
- Affordable
- Motorized Electrically

QFD Score: 1582



Engineering Modeling

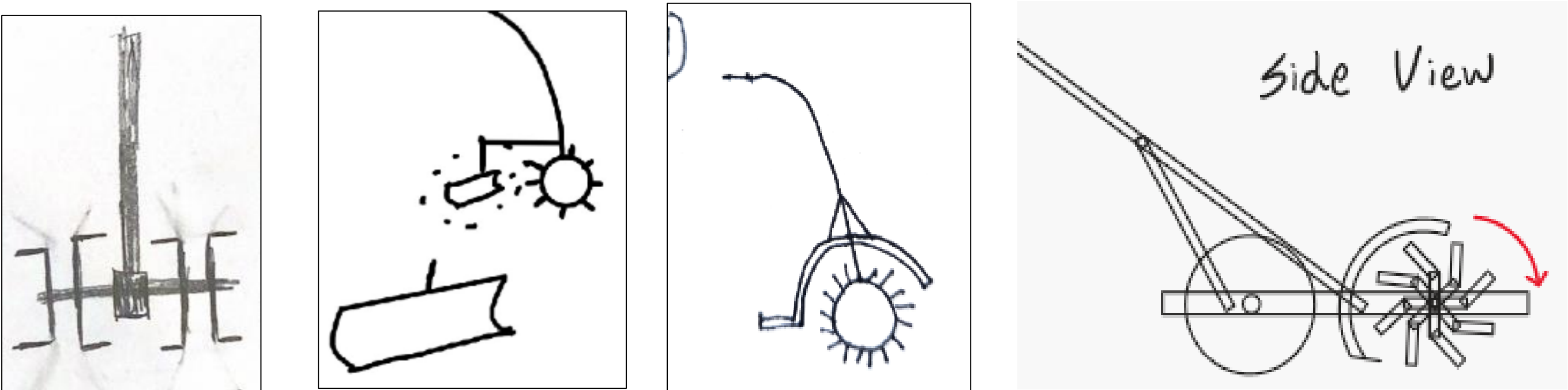


Function Tree

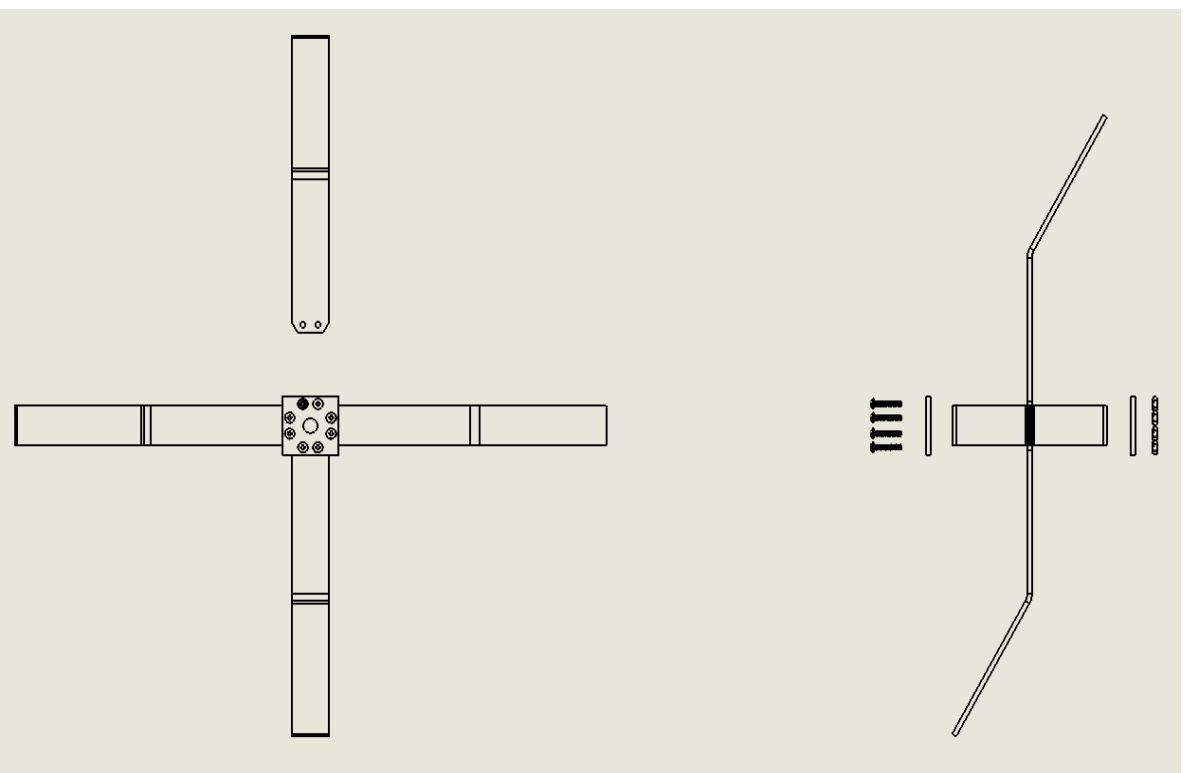
Formulas: $\omega = (v / 2\pi r)(60 \text{ sec})(GR)$, $F(p) = (30 \text{ kg} * 9.81 \text{ m/s}) * 0.08 + 50 \text{ N}$

Concept Development

Horizontally Bent plate for Guard around blades QFD Ranking: 1636
Bent Blades Leveling soil blades



Prototype Design and Fabrication

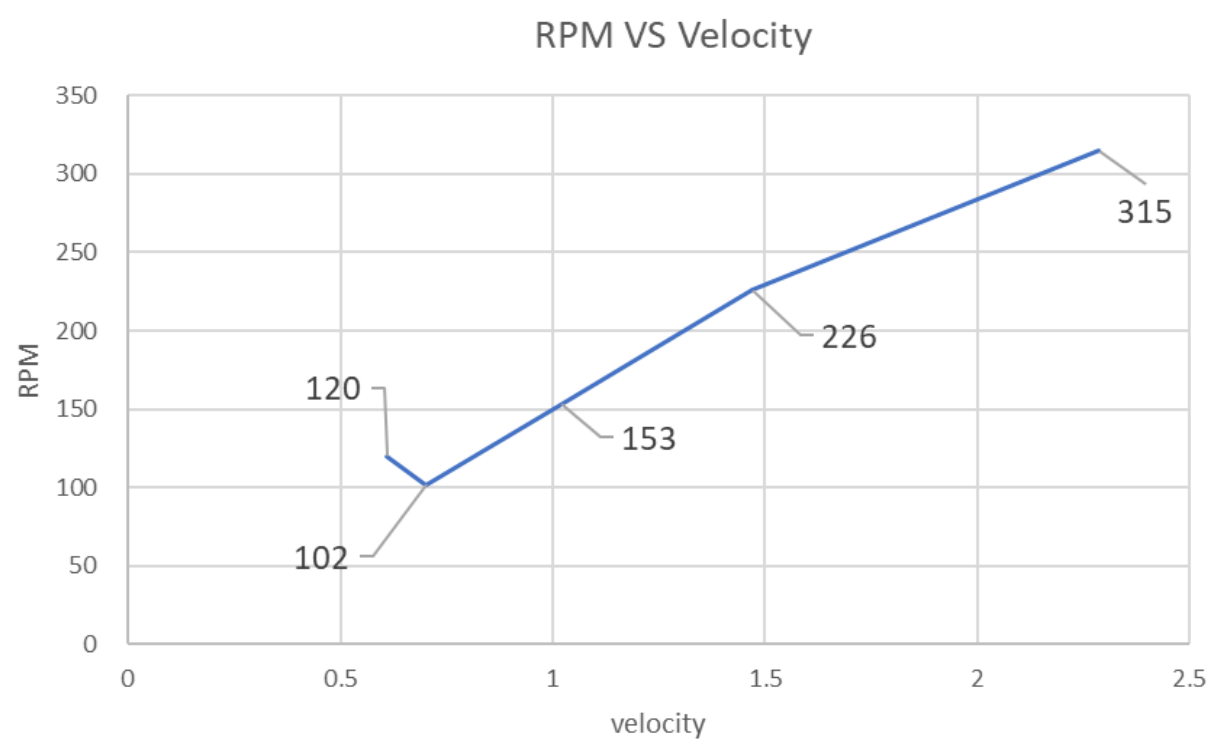


An exploded view drawing of blade assembly

Challenges Faced:

- Tolerances not tight enough for blade assembly
- Fabricating and Bending the blades
- Wooden Arm assembly strength
- Frame alignment
- Pulley tension

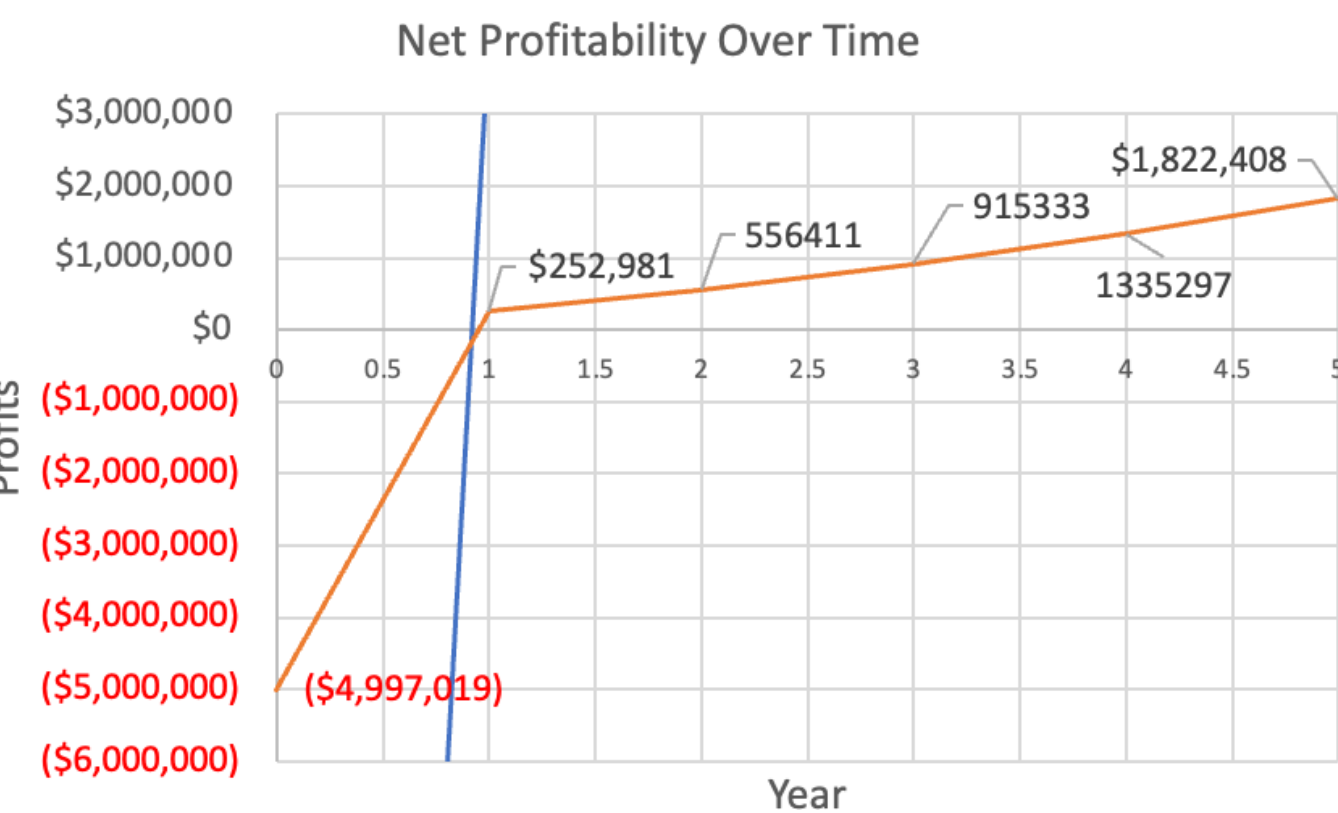
Prototype Testing



Test 1 Graph

- Test 1 RPM of blades
- Test 2 Force required to push (49N- >.401m/s)
- Test 3 Force required to reach certain depths (54N at 3.2in)

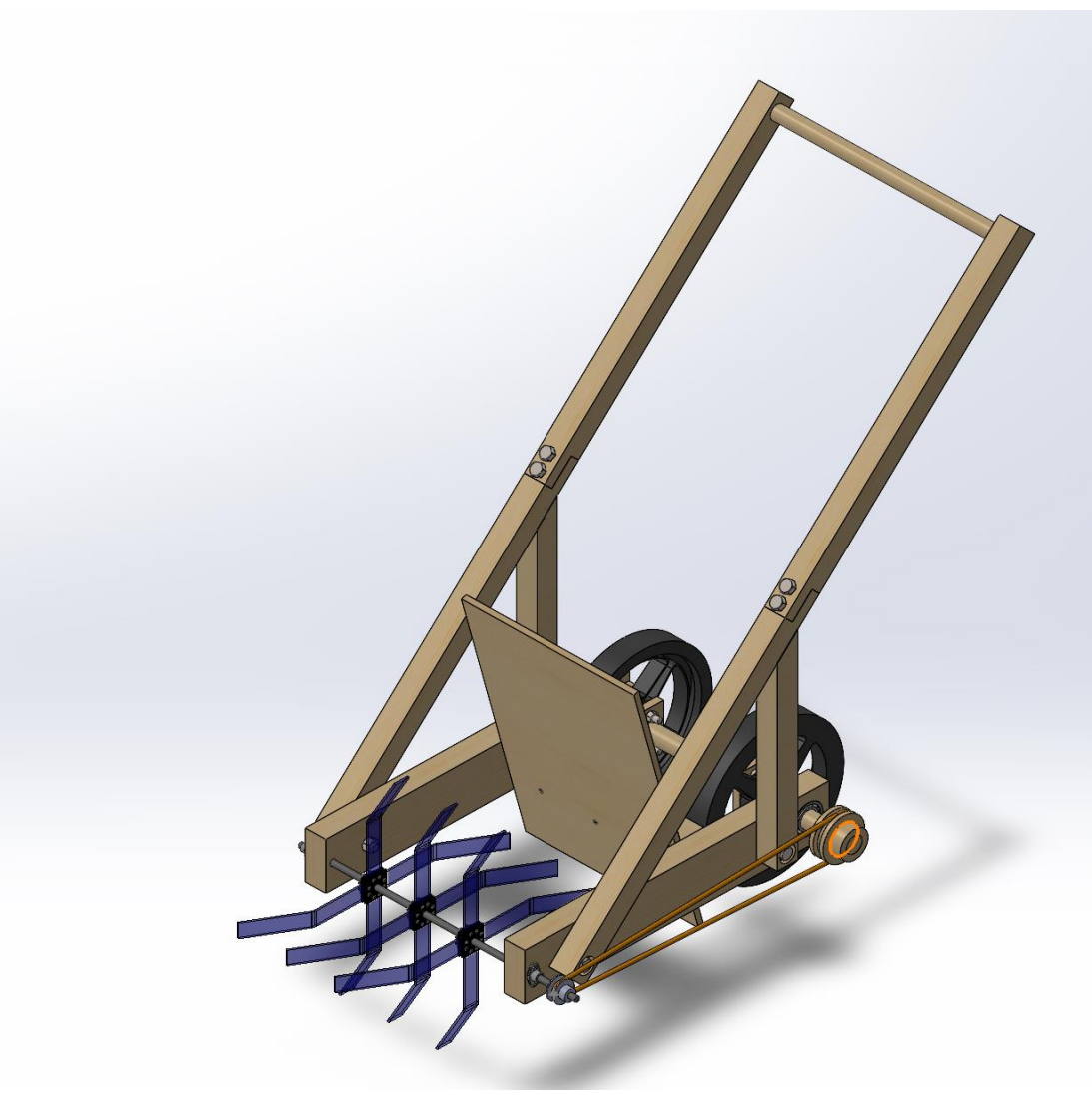
Profitability Analysis



Profitability Graph

- Employee pay per year \$3570 (104% of the poverty line)
- Market size 35,000 people
- Year 1 Profit \$252,000

Final Design



SolidWorks Model of Tiller

Changes:

- Flat belt pulley or chain + sprocket
- Aluminum square tubing for connecting arms
- Switch to wheels with better grip